RAG Evaluation Summary

A practical guide to evaluating retrieval, grounded generation, and robustness

A useful mental model

> Evaluate the retriever and generator separately first, then evaluate the complete RAG pipeline end to end. A correct answer can hide poor retrieval, while excellent retrieval can still lead to an unsupported answer.

EM means Exact Match

EM gives 1 when the prediction exactly matches the reference answer after normalization; otherwise it gives 0. The dataset score is the average, usually reported as a percentage.

Reference	Prediction
"Paris"	**"Paris" -> EM = 1**

Limitation: "The answer is Paris" may score 0 under strict EM even though it is semantically correct. EM is best for short answers with one expected wording, not open-ended LLM responses.

Three quality dimensions | Four robustness capabilities | Five evaluation frameworks

What should a RAG system evaluate?

Core quality dimensions and stress-test capabilities from the document

1. Core quality dimensions

Dimension	Question it answers	Typical metrics
Context relevance	Did retrieval return information useful for the question?	Precision, Recall, Hit Rate, MRR, NDCG
Faithfulness / fidelity	Is the answer supported by the retrieved context, without invented claims?	Faithfulness score; claim-support checks
Answer relevance	Does the generated answer directly address the user question?	Cosine similarity; relevance judge

2. Stress-test capabilities

Capability	What to test	Metrics in the document
Noise robustness	Ignore irrelevant or misleading retrieved passages	Accuracy, Precision
Negative rejection / refusal	Refuse when the supplied evidence cannot answer the question	Accuracy, EM
Information integration	Combine evidence from multiple passages correctly	Accuracy
Counterfactual robustness	Resist false statements in the supplied context	Accuracy, R-Rate

Frameworks and metric selection

Benchmarks and automated evaluation tools mentioned in the document

Framework	Target	Main aspects	Metrics / scoring
RGB	Retrieval + generation	Noise; refusal; integration; counterfactual robustness	Accuracy, EM
RECALL	Generation	Counterfactual robustness	R-Rate
RAGAS	Retrieval + generation	Context relevance; faithfulness; answer relevance	Judge scores; cosine similarity
ARES	Retrieval + generation	Context relevance; faithfulness; answer relevance	Accuracy
TruLens	Retrieval + generation	Context relevance; faithfulness; answer relevance	Feedback functions / judge scores

How to choose metrics

Retriever

Use Hit Rate or Recall to check whether useful evidence was found. Use MRR or NDCG when ranking order matters.

Generator

Use faithfulness for grounding and answer relevance to check whether the response addresses the question.

End-to-end

Add accuracy, refusal, noisy-context, and counterfactual tests. Use EM only when one exact answer is expected.

Source: "Intro of Retrieval Augmented Generation (RAG) and application demos," slides 17-19.
"Answer fidelity" and "faithfulness" refer to the same grounding concept in this summary.